

look close together only because they lie in the same line of sight from Earth. See also *binary star*

Dwarf planet

A rounded orbiting body similar to a planet but not massive enough to have cleared its orbital path of other objects. Pluto is an example. See also *planet*

E

Eclipsing binary

A binary star system in which each star passes alternately in front of the other, cutting off all or part of its light and causing a periodic variation in the system's overall brightness.

Ecliptic

(1) The track along which the Sun appears to travel around the celestial sphere, relative to the background stars, in the course of a year. (2) The plane of Earth's orbit around the Sun (which determines the position of the ecliptic in sense 1). See also *celestial sphere*, *zodiac*

Electromagnetic radiation

Radiation that transmits energy throughout the Universe as waves of fluctuating electric and magnetic fields which all travel at "the speed of light." See also *electromagnetic spectrum*

Electromagnetic spectrum

The whole range of electromagnetic radiation, from radio waves (which have the lowest frequencies and the longest wavelengths) through microwaves, infrared radiation, visible light, ultraviolet radiation, and X-rays, to gamma rays (with highest frequencies and energies, and shortest wavelengths).

Electron

A subatomic particle with a negative electric charge, found in all atoms. Electrons are much lighter than the protons and neutrons that make up atomic nuclei. See also *atom*

Elliptical galaxy

A galaxy that is elliptical or round in shape. Elliptical galaxies typically contain older stars and show little evidence of current star creation.

ESA

Short for European Space Agency, an organization supported by most European countries, with headquarters in Paris.

Exoplanet

See *extrasolar planet*

Extrasolar planet (exoplanet)

A planet orbiting a star other than the Sun.

Extremophile

Any life form that thrives under extreme conditions, such as high pressure, very high or low temperatures, or unusual chemical environments.

F

Fusion (nuclear fusion)

The process whereby atomic nuclei join to form heavier atomic nuclei at high temperatures. Stars are powered by fusion reactions that take place in their cores and release large amounts of energy.

G

Galactic plane

The plane of the flat disk of a galaxy, especially the Milky Way, where most of its stars are found.

Galaxy

A huge aggregation of star systems, gases, dust, and dark matter, held together by gravity and distinct from other surrounding galaxies. Galaxies can hold from millions up to trillions of stars. The Milky Way Galaxy is often referred to as "The Galaxy" with a capital letter. See also *active galaxy*, *elliptical galaxy*, *irregular galaxy*, *lenticular galaxy*, *spiral galaxy*

Galaxy cluster

An aggregation of 50 to 1,000 galaxies held together by gravity.

Galaxy supercluster

A cluster of galaxy clusters. A supercluster may contain 10,000 or more galaxies, spread through a volume of space with a diameter of up to about 200 million light-years.

Gamma radiation

Electromagnetic radiation of extremely short wavelengths and high frequencies and energy. See also *electromagnetic radiation*, *electromagnetic spectrum*

Gas giant

A large planet composed mainly of hydrogen and helium. Jupiter and Saturn are examples in our Solar System. See also *rocky planet*

Globular cluster

A near-spherical cluster of between 10,000 and more than 1 million stars. Globular clusters consist of very old stars and are located mainly in the spherical halo regions around galaxies.

Gravitationally bound

Phrase applied to any astronomical system that is kept together by gravitational attraction between its parts. The Solar System and the Milky Way are examples.

Gravity

An attractive force between all objects that have mass or energy, experienced on Earth as weight. The force of gravity keeps planets in orbit around the Sun and stars in orbit around the Galaxy.

H

Hertzsprung-Russell (HR) diagram

A diagram where stars are plotted according to their luminosity and surface temperature/color. See also *luminosity*, *main-sequence star*

Hot Jupiter

An extrasolar planet similar to Jupiter in size and composition, but orbiting much closer to its star and therefore hotter. See also *extrasolar planet*

Hubble constant

A mathematical constant that relates a galaxy's distance to the speed that it is receding from our own galaxy. It represents an estimate of the expansion rate of the Universe.

Hypergiant star

A star of exceptionally high mass, larger than a supergiant. Hypergiants may be 100 times or more the mass of the Sun, but are short-lived, burning themselves out quickly.

I

Infrared radiation

Electromagnetic radiation with wavelengths longer than visible light but shorter than microwaves, experienced as heat radiation in everyday life. See also *electromagnetic radiation*, *electromagnetic spectrum*

Interferometry

A technique involving measuring the overlap between electromagnetic waves from a distant source, used to achieve sharper images of an object. Information from arrays of telescopes or radio telescopes many miles apart can be combined, resulting in images approximating those from an imaginary huge telescope the size of the array.

Ion

An atom that has lost or gained one or more electrons and therefore has an overall positive or negative charge. The process of this happening is called ionization. See also *electron*

Irregular galaxy

A galaxy that lacks a well defined structure or symmetry.

K

Kuiper Belt

A region of the Solar System beyond Neptune containing bodies of icy and rocky composition. See also *Oort cloud*

L

Lenticular galaxy

A galaxy shaped like a convex lens. It has a central bulge that merges into a flattened disk, but has no spiral arms.

Light-year

A unit of measurement, defined as the distance that light travels through a vacuum in one year. 1 light-year is equal to 5,878 billion miles (9,460 billion km).

Local Group

A small cluster of over 50 galaxies that includes our own galaxy, the Milky Way. The group also contains two other large spiral galaxies, including the well-known Andromeda Galaxy, although most of its members are small elliptical or irregular galaxies. See also *galaxy cluster*

Look-back distance

The distance that light has traveled from a distant object to reach us today. It is farther than the original distance to the object (since the Universe has expanded while the light was traveling) but less than its present distance (since the object is now farther away than when it sent the light).

Luminosity

The total amount of energy emitted in one second by a source of radiation, such as the Sun or a star. See also *absolute magnitude*.

M

Magnetic field

The region around a magnetized body within which magnetic forces affect the motion of electrically charged particles.

Magnitude

A measure of the brightness of a star or other astronomical object. See also *absolute magnitude*, *apparent magnitude*

Main-sequence star

Any star that falls within the main diagonal band on a Hertzsprung-Russell diagram, in which luminosity is plotted against temperature. Main-sequence stars are converting hydrogen in their cores into helium, and may stay in the same position on the sequence for billions of years, the exact position depending mainly on the star's original mass. The Sun is a main-sequence star. See also *Hertzsprung-Russell diagram*.